

Sardar Vallabhbhai National Institute of Technology, Surat
Department of Computer Science and Engineering
M.Tech(CSE/DS) – I Sem – 2026–27
Computer Vision and Image Processing (CSCS111/CSDS119)
Lab Assignment 7 – Radiometry and Image Formation

Instructions: Use Python, MATLAB, or any suitable programming environment. Implement the required simulations, generate appropriate visualizations, and provide a brief observation based on the obtained results.

PART A: Radiometry and Image Formation

1. Develop a program to investigate how the received illumination changes with surface area and distance from a radiation source. Perform experiments using multiple values of area and distance and visualize the results.
2. Develop a numerical experiment to investigate the effect of illumination power, distance, surface orientation, and surface properties on the intensity received by a surface. Vary one parameter at a time and analyze the resulting changes.
3. Simulate two surfaces with different reflection characteristics. Change the illumination direction and viewing direction and visualize the resulting reflected intensity. Compare the appearance of the two surfaces.
4. Create a simulation for a diffuse surface and investigate how changes in illumination level and surface reflectance affect the outgoing intensity. Generate suitable plots and analyze the relationship between the parameters.
5. Develop a simplified camera image-formation simulation. Investigate how changes in camera parameters such as focal length, aperture, and exposure affect the intensity recorded at the image plane.
6. Develop a complete simplified simulation of the image-formation process from illumination to final pixel intensity. Use at least two surfaces with different reflectance properties and investigate the effect of changing illumination and surface properties.
7. Generate synthetic images of the same scene under different illumination conditions. Compare the resulting images using intensity profiles and histograms. Identify and explain the regions where the intensity changes significantly.

PART B: Experimental Investigation

Capture or obtain images of the same object under two substantially different illumination conditions, such as direct illumination and shadow.

Analyze and compare the images in terms of:

- (a) overall brightness,
- (b) pixel-intensity distribution,
- (c) mean and standard deviation of intensity,
- (d) intensity variation across selected regions, and
- (e) visible changes in surface appearance.

Relate the observed differences to the image-formation process and identify the factors responsible for the changes in the recorded image.